

PRODUCT SPECIFICATION DOCUMENT: ADVANCED STOCK TRADING & ANALYTICS APP

1. Overview

This document outlines the functional and technical specifications for an advanced stock trading and analytics platform. The platform is designed for individual investors and traders who need real-time insights, automated trading capabilities, and comprehensive financial data visualization. Features are categorized as **Free** or **Paid Requirements** based on accessibility tiers.

2. Key Features

2.1 Market Analysis (Free Requirements)

- Display Top 5 Gainers and Losers (daily)
- Show daily and % changes in stock prices
- Show price-to-volume ratio
- Filter and choose exchanges by continent, country, and index

2.2 Trade Execution (Free Requirements)

- Manual and automatic trading options
- Trade top/bottom 5 stocks using equal or percentage dollar allocation
- Support for US and selected international exchanges
- Integration with brokers like Alpaca, Interactive Brokers, TD Ameritrade

2.3 Portfolio Management (Free Requirements)

- Create and manage watchlists
- Real-time tracking of selected stocks
- Allocation visualizer and optimizer

2.4 Visualization (Free Requirements)

- Graphs with timeframes: 1D, 5D, 1M, 6M, YTD, 1Y, 5Y
- Include data overlays like volume, moving averages, Bollinger bands

2.5 Metrics & Fundamentals (Free Requirements)

- Display Open, High, Low, Close, Prev Close
- 52-week highs/lows with dates
- Beta, Market Cap, Shares Outstanding
- Dividends and YTD % change
- EPS, P/E, Revenue, ROE, EBITDA, Gross Margin, Net Margin, Debt/Equity
- Prev. Day Close, Day Close, Change Amount, % Change

2.6 News & Intelligence (Free Requirements)

- Company profiles, earnings, financials
- Real-time financial news
- Peer and sector comparisons
- Options and ownership data
- CNBC-style UI elements

2.7 Reporting (Free Requirement)

- Downloadable reports (PDF, Excel)
- Customizable reports by time range and stock list

2.8 Paid Tier Enhancements

- **Mock Trading:** Simulate top 5/bottom 5 strategies with full trade support

- **Global Trading:** Access exchanges and indices beyond the U.S.
- **Mock Funding:** Virtual dollar allocation for learning and simulation
- **Social Sentiment Aggregation:** Pull stock sentiment from social platforms
- **Voice Trading:** Trade via voice and confirm by audio interaction
- **Trade History:** View historical gains/losses in amount and %
- **Sentiment Mapping:** Match Reddit/Twitter sentiment to trade triggers
- **Family/Group Portfolio Sharing:** Dashboards with collaborative access
- **AI-Powered Back testing:** Auto-analyze trade ideas against historical data
- **Back test Comparison Mode:** Compare trade vs. hold outcomes on selected days
- **Transparency Panel:** Show which data sources influenced trade logic
- **Education Mode:** Turn trade decisions into real-time learning modules

3. Technical Architecture

3.1 Frontend

- **Framework:** React (Web), React Native (Mobile)
- **Libraries:** Chart.js, TradingView, TailwindCSS
- **State Management:** Redux / Zustand

3.2 Backend

- **Language:** Node.js or Python (FastAPI)
- **DB:** PostgreSQL, Redis (caching)
- **APIs:** RESTful and WebSocket APIs
- **Scheduler:** Celery (Python) or Bull (Node.js) for job automation

3.3 Integrations

- **Market Data:** Polygon.io, IEX Cloud, Alpha Vantage
- **News:** Benzinga, CNBC, Yahoo Finance
- **Brokerage APIs:** Alpaca, Interactive Brokers, Robinhood, TD Ameritrade
- **Social Sentiment:** Twitter API, Reddit API
- **Voice Recognition:** Google Cloud Speech or Whisper AI

3.4 Security

- OAuth2.0 and JWT for authentication
- Role-based access
- IP rate limiting

4. Roadmap

Phase	Features	Timeframe
MVP	Watchlist, Manual Trade, Top Movers, Graphs	6-8 weeks
Phase 2	Auto Trade, Reporting, CNBC UI, Ratios	4-6 weeks
Phase 3	News, Alerts, Earnings, Ownership	6-8 weeks
Phase 4	AI Insights, Multi-platform Broker Support, Sentiment Analysis, Mock Trading	8+ weeks

5. Appendix

5.1 API Contract Samples

GET /api/stocks/top-gainers

Response: {
 "date": "2025-05-16",
 "data": [
 {"symbol": "AAPL", "change": 4.52, "percent": 2.5},

```
    {"symbol": "TSLA", "change": 10.20, "percent": 3.8},  
    ...  
  ]  
}
```

POST /api/trade/execute

Request: {

```
  "symbol": "AAPL",  
  "action": "buy",  
  "amount_usd": 1000,  
  "auto_trade": true  
}
```

Response: {

```
  "status": "success",  
  "trade_id": "TRX12345678"  
}
```

5.2 User Journey Flows

- User registers and links brokerage account
- User creates watchlist from top movers or manual search
- User enables auto-trade with percentage allocation settings
- User views real-time charts and news
- User downloads weekly performance report
- User initiates mock trade with AI-driven sentiment analysis
- User backtests a trading strategy with alternate scenarios

5.3 Database Schema Outline

- **Users:** id, name, email, auth_token
- **Watchlist:** user_id, symbol, created_at
- **Trades:** id, user_id, symbol, action, amount, type, created_at
- **MarketData:** symbol, date, open, high, low, close, volume
- **News:** id, symbol, headline, sentiment, source, date
- **Sentiment:** symbol, score, source, timestamp

5.4 Wireframe Sketches

- **Dashboard View:** News panel, Top Movers, Real-Time Chart, Trade Button
- **Watchlist View:** List with price/volume and quick-trade buttons
- **Trading Modal:** Symbol input, buy/sell toggle, amount field, confirm
- **Reports View:** Calendar, filters, download buttons
- **Mock Trading Panel:** Trade history, performance, education tips
- **Transparency Panel:** Visual source map for trade decision
- **Backtest View:** Calendar, alternate trade path chart